Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001

APH401S – ALLIED PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Write down the expression for the velocity of transverse vibrations of a stretched string.
2. What is Piezo electric effect?
3. Give an expression for the energy stored in a charged capacitor.
4. State the principle of potentiometer.
5. What are Newton's rings?
6. How an air wedge is formed?
7. What do you mean by time dilation?
8. State Heisenberg's uncertainty principle.
9. What is a PN junction diode?
10. Draw the circuit diagram for NOT gate using a transistor.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Derive an expression for the velocity of sound.

(or)

b) Give the theory of production of ultrasonic waves.

12. a) Determine the growth of a charge in a RC circuit.

(or)

b) Explain the internal resistance of a cell using a potentiometer.

13. a) How the radius of curvature of a lens is determined using Newton's rings?

(or)

b) Give the theory of air wedge.

14. a) Obtain Lorentz transformation equations.

(or)

b) Derive Einstein's mass energy relation.

15. a) Explain how a zener diode acts as voltage regulator.

(or)

b) State and prove any one Demorgan's theorem.

SECTION - C (3X10=30)

Answer any **THREE** Questions

17. Explain the theory of A.C frequency using sonometer.

18. Derive an expression for growth and decay of charge in LR circuit.

19. Describe the theory of plane transmission grating.

20. Calculate the values of the energy of a particle in a one dimensional box.

21. Discuss the transistor characteristics of a transistor in CE mode.

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